

Introduction

Background

Pregnant individuals lack a reliable digital platform to track health education, identify trustworthy community resources, and communicate promptly with healthcare providers about symptoms or discomforts, leading to disengagement and delayed care.

Two Happy Hearts

A digital maternal health platform that consolidates education, surveys, alerts, and secure communication for pregnant individuals facing hypertension risks.

Overview

We built a **React Native app** for patients, a **web dashboard** for care teams, and a **Spring Boot backend** for authentication and data management. Patients can access educational resources and complete surveys, while care teams can track engagement and responses in real time.

Care Team Challenges

Nurses rely on spreadsheets, email, and manual tracking. Data is fragmented, which makes it hard to identify trends and support patients consistently.

Measurable Organizational Value (MOV)

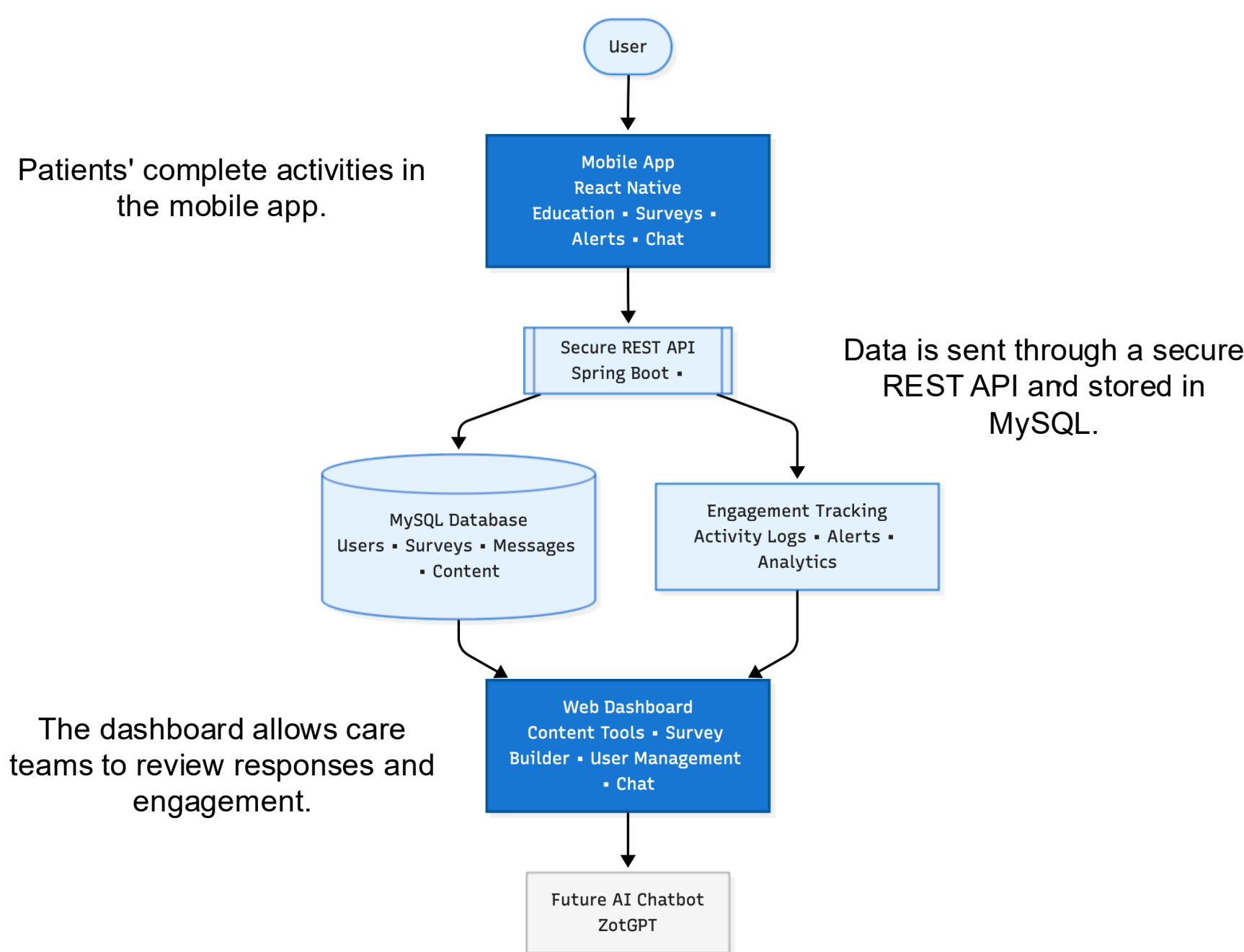
The project aims to reach 75 percent weekly engagement, unify all tools into one system, provide real-time insight into patient activity, and reduce the manual tracking required by care teams.

System Components

- **Mobile App (React Native)**
 - Education, surveys, alerts, secure chat
- **Web Dashboard (React)**
 - Content tools, survey builder, user management, engagement data, secure chat
- **Backend API (Spring Boot)**
 - Secure login, REST endpoints, data processing
- **Database (MySQL)**
 - Stores users, surveys, messages, educational content

Architecture

A simplified diagram below shows how the app, backend, and dashboard work together.



Key Achievements

- Completed full **mobile** → **backend** → **dashboard** data pipeline
- Implemented secure **JWT authentication**
- Delivered working **education, survey, engagement, and chat** modules
- Deployed **TestFlight mobile build** and **live dashboard web app**
- Added **alerting** and **analytics** to support timely patient monitoring

Mobile App Updates

- **New login screen** with improved layout and accessibility
- **Redesigned home page** showing sleep, readiness, and blood pressure metrics
- **Improved survey flow** for daily check-ins
- **Streamlined chat interface** for clearer communication

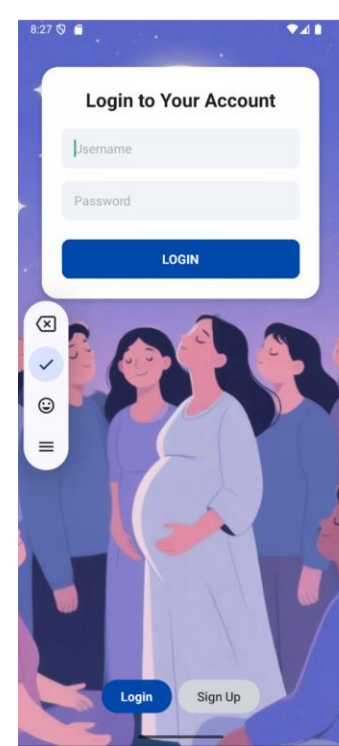


Figure 1.
Login screen

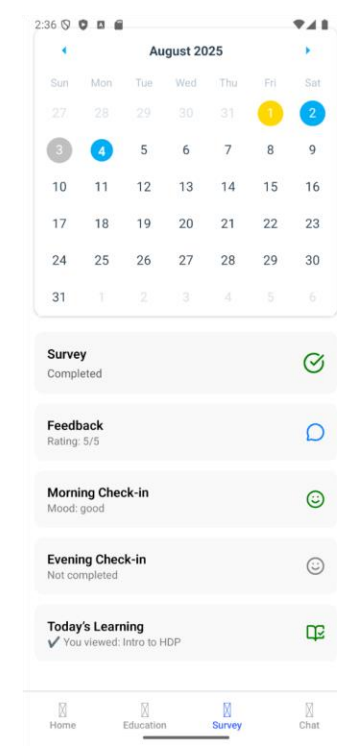


Figure 2.
Survey calendar
and check-in flow

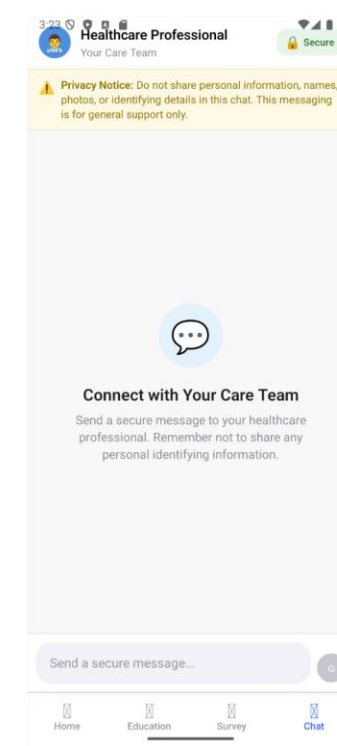


Figure 3.
Chat interface for
secure
communication

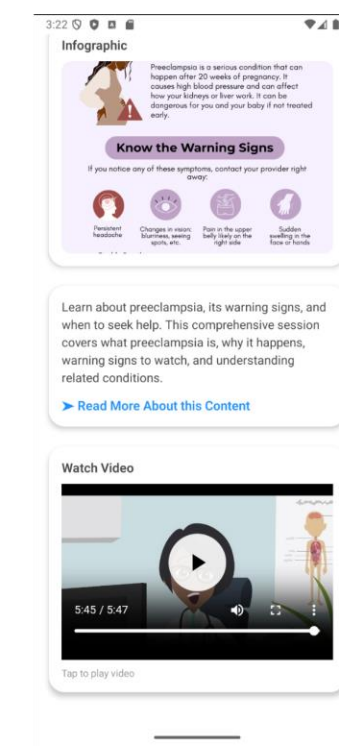


Figure 4.
Education module
with visuals and
video

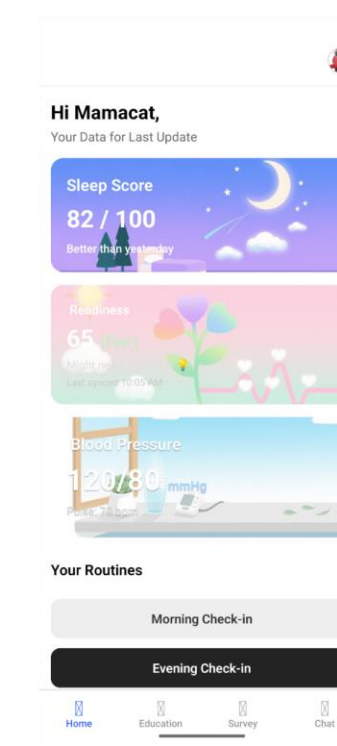


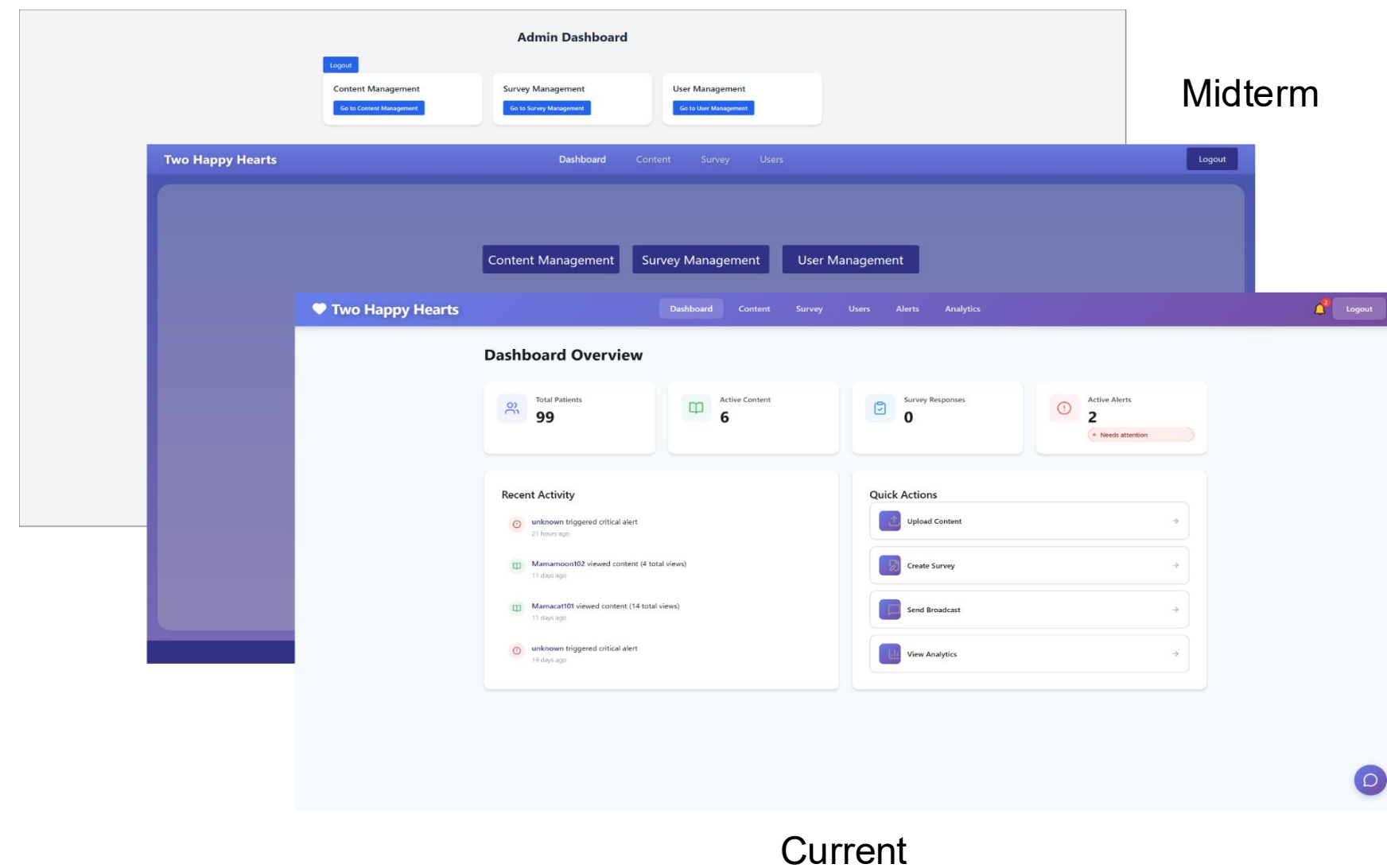
Figure 5.
Home page with
wellness
metrics

Dashboard Updates

- Updated **main page** with relevant information and easier navigation
- Enhanced **user management** view for tracking for tracking participants and activity
- Designed **survey management** with sortable submissions and clear status
- Organized **education management** with updated editing tools and categories
- Improved **chat interface** with secure, threaded messaging
- **Created automated alert system** that flags urgent or concerning survey responses
- **Developed analytics panel** that tracks what educational content participants have viewed

Visual Updates

Original



Current

What's Next?

- Migration from Railway to Amazon Web Services (AWS).
- User Management system improvements
- Content and Survey Management improvements
- Patient-interfacing AI agent utilizing UCI's ZotGPT API.
- Interactive calendar feature